Raw Material Procurement-to-Payment Data Dictionary

Procurement

1. Supplier Master

Column	Data Type	NULL?	Default	Purpose
supplier_id	BIGINT	NO	Identity	Primary key
supplier_code	VARCHAR(30)	NO	—	ERP supplier number
supplier_name	VARCHAR(200)	NO	—	Supplier/company name
supplier_type_id	BIGINT	NO	—	FK to supplier type
business_registration_no	VARCHAR(100)	YES	—	Business registration
tin_no	VARCHAR(50)	YES	—	Tax Identification Number
vat_no	VARCHAR(50)	YES	—	VAT registration
company_description	TEXT	YES	—	Description
registration_date	DATE	NO	Current date	ERP registration date
status	VARCHAR(20)	NO	ACTIVE	Supplier status
created_by	BIGINT	NO	—	User who created record
created_at	TIMESTAMPTZ	NO	Current timestamp	Creation time
updated_by	BIGINT	YES	—	Last modifying user
updated_at	TIMESTAMPTZ	YES	—	Last modification time
operating_unit_id	INT	NO	-	Operating unit

2. Supplier Contact

Field	Data Type	NULL?	Default	Description
supplier_contact_id	BIGINT	NO	Identity	Primary key
supplier_id	BIGINT	NO	—	FK to supplier
contact_name	VARCHAR(150)	NO	—	Contact person name
position	VARCHAR(100)	YES	—	Contact person's position
phone_number	VARCHAR(30)	YES	—	Office/landline phone
mobile_number	VARCHAR(30)	YES	—	Mobile phone
email	VARCHAR(150)	YES	—	Email address
is_primary	BOOLEAN	NO	FALSE	Indicates primary contact
status	VARCHAR(20)	NO	ACTIVE	Contact status
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time

3. Supplier Address

Field	Data Type	NULL?	Default	Description
supplier_address_id	BIGINT	NO	Identity	Primary key
supplier_id	BIGINT	NO	—	FK to supplier
address_type	VARCHAR(30)	NO	—	Type of address, e.g. PHYSICAL, MAILING, BILLING
country	VARCHAR(100)	NO	—	Country
region_state	VARCHAR(100)	YES	—	Region/State
city	VARCHAR(100)	NO	—	City
postal_code	VARCHAR(30)	YES	—	Postal/ZIP code
physical_address	TEXT	YES	—	Physical location/address
mailing_address	TEXT	YES	—	Mailing address
is_primary	BOOLEAN	NO	FALSE	Indicates primary address
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time

4. Finical Information

Field	Data Type	NULL?	Default	Description
supplier_bank_account_id	BIGINT	NO	Identity	Primary key
supplier_id	BIGINT	NO	—	FK to supplier
bank_name	VARCHAR(150)	NO	—	Bank name
branch_name	VARCHAR(150)	YES	—	Bank branch name
account_name	VARCHAR(200)	NO	—	Name registered on the bank account
account_number	VARCHAR(100)	NO	—	Bank account number
swift_code	VARCHAR(50)	YES	—	SWIFT/BIC code
currency_id	BIGINT	NO	—	FK to currency master
is_primary	BOOLEAN	NO	FALSE	Indicates primary bank account
status	VARCHAR(20)	NO	ACTIVE	Account status
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time

2. Purchase Requisition (PR)

Material Request Header

Field	Data Type	NULL?	Default	Description
material_request_id	BIGINT	NO	Identity	Primary key
request_number	VARCHAR(30)	NO	—	Unique material request number
request_date	DATE	NO	CURRENT_DATE	Date the request was created
requesting_department_id	BIGINT	NO	—	Department requesting the materials
requestor_employee_id	BIGINT	NO	—	Employee making the request
required_date	DATE	NO	—	Date materials are needed
priority_level	VARCHAR(20)	NO	MEDIUM	Low, Medium, High, Urgent
purpose	VARCHAR(100)	NO	—	Production, Maintenance, Testing, Office Use, etc.
request_status	VARCHAR(20)	NO	DRAFT	Draft, Submitted, Approved, Rejected, Fulfilled, Closed
remarks	TEXT	YES	—	Additional notes
created_by	BIGINT	NO	—	User who created the request
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	System creation date/time
operating_unit_id	INT	NO	-	Operating unit

Material Request Details (Line Items)

Field	Data Type	NULL?	Default	Description
material_request_line_id	BIGINT	NO	Identity	Primary key / unique line item identifier
material_request_id	BIGINT	NO	—	FK to material_request
item_id	BIGINT	NO	—	FK to item_master, material identifier
item_code	VARCHAR(50)	NO	—	Material/item code
item_name	VARCHAR(200)	NO	—	Material name
item_category_id	BIGINT	NO	—	FK to item category
uom_id	BIGINT	NO	—	FK to unit of measure
requested_quantity	NUMERIC(18,4)	NO	0	Quantity requested
available_stock	NUMERIC(18,4)	NO	0	Current available stock
quantity_to_purchase	NUMERIC(18,4)	NO	0	Quantity that must be procured
estimated_unit_cost	NUMERIC(18,4)	YES	—	Expected unit price
estimated_total_cost	NUMERIC(18,4)	YES	—	Quantity to purchase × estimated unit cost
specification	TEXT	YES	—	Material specifications
notes	TEXT	YES	—	Item-specific remarks
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time

PR Approval Information

Attribute	Description
Approval ID	Approval record identifier
Approver Employee ID	Manager or authorized approver
Approval Date	Date approved/rejected

Approval Status	Pending, Approved, Rejected
Approval Remarks	Comments from approver
Operating unit id	Operating unit

3. Purchase Order (Po)

Purchase Order — Header Table

Field	Data Type	NULL?	Default	Description
purchase_order_id	BIGINT	NO	Identity	Primary key
po_number	VARCHAR(30)	NO	—	Unique purchase order identifier
po_date	DATE	NO	CURRENT_DATE	Date the purchase order is created
material_request_id	BIGINT	YES	—	FK to approved material/purchase request
request_number	VARCHAR(30)	YES	—	Reference number of approved purchase request
supplier_id	BIGINT	NO	—	FK to supplier
payment_term_id	BIGINT	YES	—	FK to payment terms
delivery_term_id	BIGINT	YES	—	FK to delivery terms
purchasing_department_id	BIGINT	NO	—	Procurement department issuing the PO
buyer_employee_id	BIGINT	NO	—	Procurement officer responsible for the order
plant_id	BIGINT	NO	—	Receiving flour mill/factory location
po_status	VARCHAR(30)	NO	DRAFT	Draft, Pending Approval, Approved, Sent, Closed, Cancelled
tax_amount	NUMERIC(18,2)	NO	0	Applicable VAT or other taxes
discount_amount	NUMERIC(18,2)	NO	0	Negotiated discount
required_delivery_date	DATE	NO	—	Date material is needed
warehouse_id	BIGINT	NO	—	Receiving warehouse or storage location
created_by	BIGINT	NO	—	User who created the PO
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	System creation date/time
updated_by	BIGINT	YES	—	User who last updated the PO
updated_at	TIMESTAMPTZ	YES	—	Last update date/time
operating_unit_id	INT	NO	-	Operating unit

Purchase Order Line

Field	Data Type	NULL?	Default	Description
purchase_order_line_id	BIGINT	NO	Identity	Primary key
purchase_order_id	BIGINT	NO	—	FK to purchase_order
line_number	INTEGER	NO	—	Item sequence number
item_id	BIGINT	NO	—	FK to item_master
item_code	VARCHAR(50)	NO	—	Material or product code
uom_id	BIGINT	NO	—	FK to uom
quantity_ordered	NUMERIC(18,4)	NO	0	Quantity ordered
unit_price	NUMERIC(18,4)	NO	0	Price per unit
total_line_amount	NUMERIC(18,2)	NO	0	Quantity × Unit Price
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time

Purchase Order Approval

Field	Data Type	NULL?	Default	Description
purchase_order_approval_id	BIGINT	NO	Identity	Primary key
purchase_order_id	BIGINT	NO	—	FK to purchase_order
requestor_employee_id	BIGINT	NO	—	Employee requesting the purchase
department_id	BIGINT	NO	—	Requesting department
budget_code	VARCHAR(50)	YES	—	Cost center or budget allocation code
approval_level	INTEGER	NO	—	Approval hierarchy level
approver_employee_id	BIGINT	NO	—	Employee/person responsible for approval
approval_date	TIMESTAMPTZ	YES	—	Approval/rejection timestamp
approval_status	VARCHAR(20)	NO	PENDING	Pending, Approved, Rejected
approval_comments	TEXT	YES	—	Remarks from approver
created_at	TIMESTAMPTZ	NO	CURRENT_TIMESTAMP	Record creation date/time
updated_at	TIMESTAMPTZ	YES	—	Last update date/time
operating_unit_id	INT	NO	-	Operating unit

Inventory

1. Warehouse master

Column	Data Type	NULL?	Default	Purpose
warehouse_id	BIGSERIAL	NO	Identity	Primary key; unique system-generated warehouse ID
operating_unit_id	BIGINT	NO	—	FK to operating_units; identifies the factory/operating unit owning the warehouse
warehouse_code	VARCHAR(30)	NO	—	Unique warehouse code, e.g. WH-RM-001
warehouse_name	VARCHAR(150)	NO	—	Name of the warehouse
warehouse_type	VARCHAR(30)	NO	—	Warehouse type, e.g. Raw Material, Finished Goods, Spare Parts, General
description	TEXT	YES	—	Detailed description of the warehouse
storekeeper_id	BIGINT	YES	—	FK to employee/user responsible for the warehouse
capacity	NUMERIC(18,3)	YES	—	Maximum storage capacity
capacity_uom	VARCHAR(20)	YES	—	Capacity unit, e.g. TON, KG, PALLET, M3
current_capacity	NUMERIC(18,3)	YES	0	Current utilized storage capacity
temperature_controlled	BOOLEAN	NO	FALSE	Indicates whether temperature is controlled
humidity_controlled	BOOLEAN	NO	FALSE	Indicates whether humidity is controlled
allow_receipt	BOOLEAN	NO	TRUE	Indicates whether materials can be received into the warehouse
allow_issue	BOOLEAN	NO	TRUE	Indicates whether materials can be issued from the warehouse
allow_transfer	BOOLEAN	NO	TRUE	Indicates whether materials can be transferred to another warehouse/bin
status	VARCHAR(20)	NO	'ACTIVE'	Warehouse status, e.g. ACTIVE / INACTIVE
effective_from	DATE	YES	—	Date from which the warehouse becomes effective
effective_to	DATE	YES	—	Date on which the warehouse becomes inactive/ends
created_by	BIGINT	NO	—	User who created the warehouse record
created_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time the record was created
updated_by	BIGINT	YES	—	User who last updated the record
updated_date	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date and time of the last update
operating_unit_id	INT	NO	-	Operating unit

2. Store Master

Column	Data Type	NULL?	Default	Purpose
store_id	BIGSERIAL	NO	Identity	Primary key; unique system-generated store ID
store_code	VARCHAR(30)	NO	—	Unique store code, e.g. ST-MNT-001
store_name	VARCHAR(150)	NO	—	Name of the store
store_type	VARCHAR(30)	NO	—	Store type, e.g. Production, Maintenance, Packaging, Laboratory, General
description	TEXT	YES	—	Description of the store
warehouse_id	BIGINT	NO	—	FK to warehouses; identifies the parent warehouse
storekeeper_id	BIGINT	YES	—	FK to employee/user responsible for the store
capacity	NUMERIC(18,3)	YES	—	Maximum storage capacity
capacity_uom	VARCHAR(20)	YES	—	Capacity unit, e.g. KG, TON, M3, PALLET
current_capacity	NUMERIC(18,3)	YES	0	Current utilized storage capacity
allow_receipt	BOOLEAN	NO	TRUE	Indicates whether the store can receive materials
allow_issue	BOOLEAN	NO	TRUE	Indicates whether materials can be issued from the store
allow_transfer	BOOLEAN	NO	TRUE	Indicates whether materials can be transferred from/to the store
stock_valuation_method	VARCHAR(30)	NO	'MOVING_AVERAGE'	Inventory valuation method, e.g. FIFO or Moving Average
material_category	VARCHAR(50)	YES	—	Material categories allowed in the store
temperature_controlled	BOOLEAN	NO	FALSE	Indicates whether temperature is controlled
humidity_controlled	BOOLEAN	NO	FALSE	Indicates whether humidity is controlled
restricted_access	BOOLEAN	NO	FALSE	Indicates whether access to the store is restricted
status	VARCHAR(20)	NO	'ACTIVE'	Store status, e.g. ACTIVE / INACTIVE
effective_from	DATE	YES	—	Date from which the store becomes effective
effective_to	DATE	YES	—	Date on which the store becomes inactive
created_by	BIGINT	NO	—	User who created the store record
created_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time the record was created
updated_by	BIGINT	YES	—	User who last updated the record

updated_date	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date and time of the last update
operating_unit_id	INT	NO	-	Operating unit

3. Silo master

Column	Data Type	NULL?	Default	Purpose
silo_id	BIGSERIAL	NO	Identity	Primary key; unique system-generated silo ID
silo_code	VARCHAR(30)	NO	—	Unique silo code, e.g. SILO-W-001
silo_name	VARCHAR(150)	NO	—	Name of the silo
silo_type	VARCHAR(30)	NO	—	Silo type/material purpose, e.g. Wheat, Flour, Bran, Additive
description	TEXT	YES	—	Detailed description of the silo
warehouse_id	BIGINT	NO	—	FK to warehouses; identifies the parent warehouse
maximum_capacity	NUMERIC(18,3)	NO	—	Maximum physical storage capacity
capacity_uom	VARCHAR(20)	NO	—	Capacity unit, e.g. TON, KG, M3
minimum_level	NUMERIC(18,3)	YES	0	Minimum allowable operating level
maximum_level	NUMERIC(18,3)	YES	—	Maximum recommended operating level
safety_stock	NUMERIC(18,3)	YES	0	Minimum quantity that should normally be maintained
current_level	NUMERIC(18,3)	YES	0	Current quantity stored in the silo
status	VARCHAR(20)	NO	'ACTIVE'	Silo status, e.g. ACTIVE, INACTIVE, MAINTENANCE
created_by	BIGINT	NO	—	User who created the silo record
created_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time the record was created
updated_by	BIGINT	YES	—	User who last updated the silo
updated_date	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date and time of the last update
operating_unit_id	INT	NO	-	Operating unit

4. Bin Master

Column	Data Type	NULL?	Default	Purpose
bin_id	BIGSERIAL	NO	Identity	Primary key; unique system-generated bin ID
bin_code	VARCHAR(30)	NO	—	Unique bin code, e.g. BIN-WHT-01
bin_name	VARCHAR(100)	NO	—	Human-readable name of the bin
warehouse_id	BIGINT	NO	—	FK to warehouses; identifies the warehouse where the bin belongs
location_id	BIGINT	YES	—	FK to storage location; identifies the specific storage area/location
bin_type_id	BIGINT	YES	—	FK to bin type; identifies the bin classification/category
capacity_qty	NUMERIC(18,3)	NO	—	Maximum quantity the bin can hold
capacity_uom_id	BIGINT	NO	—	FK to UOM; identifies the unit used for bin capacity
allow_mixed_items	BOOLEAN	NO	FALSE	Indicates whether different items/materials can be stored in the same bin
allow_mixed_batches	BOOLEAN	NO	FALSE	Indicates whether different lots/batches of the same item can share the bin
temperature_min	NUMERIC(8,2)	YES	—	Minimum permitted storage temperature
temperature_max	NUMERIC(8,2)	YES	—	Maximum permitted storage temperature
status	VARCHAR(20)	NO	'ACTIVE'	Bin status, e.g. ACTIVE, BLOCKED, INACTIVE
description	TEXT	YES	—	Additional information about the bin
created_by	BIGINT	NO	—	User who created the bin record
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time the bin was created
updated_by	BIGINT	YES	—	User who last modified the bin
updated_at	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date and time of the last modification
operating_unit_id	INT	NO	-	Operating unit

2. Item Master

Column	Data Type	NULL?	Default	Purpose
material_id	BIGSERIAL	NO	Identity	Primary key; unique identifier for each material/item
material_code	VARCHAR(30)	NO	—	Unique ERP material/item code
material_name	VARCHAR(200)	NO	—	Name of the material/item

material_category	VARCHAR(50)	NO	—	Main category, e.g. Wheat, Additive, Packaging, Stationery
material_type	VARCHAR(100)	YES	—	Specific type/grade, e.g. Hard Wheat, Soft Wheat, Durum, Bag
uom	VARCHAR(20)	NO	—	Unit of measure, e.g. KG, TON, BAG, PCS
standard_cost	NUMERIC(18,4)	YES	0	Standard cost/unit used for costing and valuation
reorder_level	NUMERIC(18,3)	YES	0	Minimum stock level that triggers replenishment/PR consideration
maximum_level	NUMERIC(18,3)	YES	0	Maximum desired stock level
safety_stock	NUMERIC(18,3)	YES	0	Minimum buffer quantity maintained to protect production from stock shortages
status	VARCHAR(20)	NO	'ACTIVE'	Item status, e.g. ACTIVE or INACTIVE
inspection_required	BOOLEAN	NO	FALSE	Indicates whether the material requires QC inspection during receiving
is_raw_material	BOOLEAN	NO	FALSE	Indicates whether the item is a raw material
operating_unit_id	INT	NO	-	Operating unit

6. Supplier Delivery

Delivery Header

Column	Data Type	NULL?	Default	Purpose
supplier_delivery_id	BIGSERIAL	NO	Identity	Primary key; unique supplier delivery record
supplier_id	BIGINT	NO	—	FK to suppliers; identifies the supplier making the delivery
po_id	BIGINT	NO	—	FK to purchase_orders; identifies the related purchase order
delivery_note_number	VARCHAR(50)	NO	—	Supplier's delivery note/reference number
delivery_date	DATE	NO	—	Date shown on the supplier's delivery document
truck_plate_number	VARCHAR(30)	YES	—	Vehicle/truck registration plate number
driver_name	VARCHAR(150)	YES	—	Name of the truck driver
arrival_date	TIMESTAMP	YES	—	Date and time the truck arrived at the factory
delivery_status	VARCHAR(30)	NO	'RECEIVED'	Delivery status, e.g. PENDING, ARRIVED, INSPECTION, ACCEPTED, REJECTED, COMPLETED
operating_unit_id	BIGINT	NO	—	FK to operating_units; identifies the factory receiving the delivery
created_at	TIMESTAMP	NO	CURRENT_TIME	Date and time the record was created
created_by	BIGINT	NO	—	User who created the delivery record
updated_at	TIMESTAMP	YES	CURRENT_TIME	Date and time the record was last updated
updated_by	BIGINT	YES	—	User who last updated the delivery record

Delivery Line

Column	Data Type	NULL?	Default	Purpose
delivery_line_id	BIGSERIAL	NO	Identity	Primary key; unique supplier delivery line ID
material_id	BIGINT	NO	—	FK to item_master; identifies the delivered material/item
uom	TEXT	NO	—	UOM, the unit of measure
preliminary_quantity	NUMERIC(18,3)	NO	0	Initial quantity declared/estimated at delivery before QC and final receiving
remarks	TEXT	YES	—	Line-level remarks about the delivered item
created_at	TIMESTAMP	NO	CURRENT_TIME	Date and time the record was created
created_by	BIGINT	NO	—	User who created the delivery line
updated_at	TIMESTAMP	YES	CURRENT_TIME	Date and time the record was last updated
updated_by	BIGINT	YES	—	User who last updated the delivery line

7. GRN

GRN_Header

Column	Data Type	NULL?	Default	Purpose
grn_id	BIGSERIAL	NO	Identity	Primary key; unique GRN identifier
grn_number	VARCHAR(30)	NO	—	Unique GRN number, e.g. GRN-2026-00001
supplier_id	BIGINT	NO	—	FK to Supplier; identifies the supplier
purchase_order_id	BIGINT	NO	—	FK to Purchase Order; PO reference
supplier_delivery_id	BIGINT	NO	—	FK to Supplier Delivery; delivery reference
operating_unit_id	BIGINT	NO	—	FK to Operating Unit; receiving factory
warehouse_id	BIGINT	NO	—	FK to Warehouse; receiving warehouse
receiving_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time goods were received
receiving_officer_id	BIGINT	NO	—	FK to Employee/User; officer responsible for receiving
status	VARCHAR(20)	NO	'DRAFT'	GRN status: DRAFT, APPROVED, POSTED, CANCELLED
created_by	BIGINT	NO	—	FK to User; user who created the GRN
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	GRN creation date/time
posted_by	BIGINT	YES	—	FK to User; user who posted the GRN to inventory
posted_at	TIMESTAMP	YES	—	Date/time the GRN was posted to inventory

GRN_Line

Column	PostgreSQL Data Type	NULL?	Default	Purpose
line_id	BIGSERIAL	NO	Identity	Primary key; unique GRN line identifier
grn_id	BIGINT	NO	—	FK to grns.grn_id; identifies the parent GRN
material_id	BIGINT	NO	—	FK to Item/Material Master; identifies the received item/raw material
ordered_quantity	NUMERIC(18,3)	NO	0	Quantity ordered on the Purchase Order
delivered_quantity	NUMERIC(18,3)	NO	0	Quantity physically delivered by the supplier
accepted_quantity	NUMERIC(18,3)	NO	0	Quantity accepted after receiving/QC
rejected_quantity	NUMERIC(18,3)	NO	0	Quantity rejected or not accepted
bag_quantity	NUMERIC(18,3)	YES	0	Number of bags received/accepted, where applicable
uom_id	BIGINT	NO	—	FK to UOM Master; unit for the quantities
batch_id	BIGINT	YES	—	FK to Batch/Lot Master; identifies the material batch
bin_id	BIGINT	YES	—	FK to Bin Master; physical receiving/storage bin

7. Row Inventory Transactions/ GRN/Transfer/adjustment/return/issue

Transactions —Header

Column	Data Type	NULL?	Default	Purpose
transaction_id	BIGSERIAL	NO	Identity	Primary key; unique inventory transaction ID
operating_unit_id	BIGINT	NO	—	FK to operating units; identifies the factory where the transaction occurs
transaction_no	VARCHAR(30)	NO	—	Unique inventory transaction number, e.g. GRN-2026-00001
grn_id	BIGINT	YES		When goods in GRN
issue_id	BIGINT	YES		Issue of goods
transaction_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time of the inventory transaction
adjustment_id	BIGINT			Adjustment
transaction_type	VARCHAR(30)	NO	—	Transaction type, e.g. GRN, ISSUE, TRANSFER, ADJUSTMENT, RETURN
reference_type	VARCHAR(30)	YES	—	Source document type, e.g. PO, GRN, PHYSICAL_COUNT, PRODUCTION
reference_id	BIGINT	YES	—	ID of the related source document
status	VARCHAR(20)	NO	'DRAFT'	Transaction status, e.g. DRAFT, POSTED, CANCELLED
warehouse_id	BIGINT	NO	—	FK to warehouses; warehouse affected by the transaction
remarks	TEXT	YES	—	General remarks or notes
created_by	BIGINT	NO	—	User who created the transaction
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time the transaction was created
posted_by	BIGINT	YES	—	User who posted/approved the transaction
posted_at	TIMESTAMP	YES	—	Date and time the transaction was posted
Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Updated_by	TEXT	YES		The user who updated this record

Transaction lines — Stock Ledger

Column	Data Type	NULL?	Default	Purpose
ledger_id	BIGSERIAL	NO	Identity	Primary key; unique stock ledger record
transaction_id	BIGINT	NO	—	FK to inventory_transactions; identifies the header transaction
line_no	INT	NO	—	Sequential line number within the transaction
item_id	BIGINT	NO	—	FK to item_master; identifies the material/item
warehouse_id	BIGINT	NO	—	FK to warehouses; identifies the warehouse
location_id	BIGINT	YES	—	FK to storage location/store; identifies the storage area
bin_id	BIGINT	YES	—	FK to bins; identifies the physical bin
batch_id	BIGINT	YES	—	FK to batch/lot; identifies the material batch
uom_id	BIGINT	NO	—	FK to UOM; unit of measure for the quantity
qty_in	NUMERIC(18,3)	NO	0	Quantity received/added to inventory
qty_out	NUMERIC(18,3)	NO	0	Quantity issued/removed from inventory
unit_cost	NUMERIC(18,2)	NO	0	Cost per unit
total_cost	NUMERIC(18,2)	NO	0	Total transaction cost
balance_qty	NUMERIC(18,3)	NO	0	Running inventory balance after the transaction
remarks	TEXT	YES	—	Line-level remarks or notes
created_at	TIMESTAMP	NO	CURRENT_TIME	Date and time the ledger record was created

Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Crated_by	TEXT	NO		The user who created this record
Updated_by	TEXT	YES		The user who updated this record

8. Goods transfer

Header

Column	Data Type	NULL?	Default	Purpose
transfer_id	BIGSERIAL	NO	Identity	Primary key; unique transfer transaction ID
transfer_no	VARCHAR(30)	NO	—	Unique transfer number, e.g. TRF-2026-00001
transfer_date	TIMESTAMP	NO	CURRENT_TIME	Date and time the stock transfer is initiated
reference_type	VARCHAR(30)	YES	—	Source document type, e.g. PRODUCTION, REQUISITION, ADJUSTMENT, OTHER
operating_unit_id	BIGINT	NO	-	FK to operating units; identifies the factory where the transaction occurs
reference_no	VARCHAR(50)	YES	—	Reference document number related to the transfer
warehouse_id	BIGINT	NO	—	FK to Warehouse; warehouse responsible for the transfer
source_storage_location_id	BIGINT	NO	—	FK to Storage Location/Store; location from which stock is transferred
source_bin_id	BIGINT	YES	—	FK to Bin Master; source physical bin
destination_storage_location_id	BIGINT	NO	—	FK to Storage Location/Store; destination location
destination_bin_id	BIGINT	YES	—	FK to Bin Master; destination physical bin
remarks	TEXT	YES	—	Additional comments or explanation for the transfer
status	VARCHAR(20)	NO	'DRAFT'	Transfer status: DRAFT, APPROVED, POSTED, CANCELLED
created_by	BIGINT	NO	—	FK to User; user who created the transfer
created_at	TIMESTAMP	NO	CURRENT_TIME	Date/time transfer was created
approved_by	BIGINT	YES	—	FK to User; user who approved the transfer
approved_at	TIMESTAMP	YES	—	Date/time transfer was approved
posted_by	BIGINT	YES	—	FK to User; user who posted the transfer to inventory
posted_at	TIMESTAMP	YES	—	Date/time inventory transfer was posted
Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Updated_by	TEXT	YES		The user who updated this record

Transfer Line-Item

Column	Data Type	NULL?	Default	Purpose
transfer_line_id	BIGSERIAL	NO	Identity	Primary key; unique transfer line ID
transfer_id	BIGINT	NO	—	FK to transfers.transfer_id; parent transfer
material_id	BIGINT	NO	—	FK to Item/Material Master; identifies the item being transferred
item_code	VARCHAR(30)	NO	—	Item/material code; normally obtained from Item Master
item_name	VARCHAR(200)	NO	—	Item/material name; normally obtained from Item Master
batch_id	BIGINT	YES	—	FK to Batch/Lot Master; identifies the batch being transferred
batch_no	VARCHAR(50)	YES	—	Batch/Lot number for traceability
uom_id	BIGINT	NO	—	FK to UOM Master
available_qty	NUMERIC(18,3)	NO	0	Quantity available at the source location/bin before transfer
transfer_qty	NUMERIC(18,3)	NO	0	Quantity to be transferred
remarks	TEXT	YES	—	Line-specific comments or notes
created_by	BIGINT	NO	—	FK to User; user who created the transfer
created_at	TIMESTAMP	NO	CURRENT_TIME	Date/time transfer was created
Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Updated_by	TEXT	YES		The user who updated this record

9. Production Material Issue

Material Issue Header

Column	Data Type	NULL?	Default	Purpose
issue_id	BIGSERIAL	NO	Identity	Primary key; unique material issue ID
issue_number	VARCHAR(30)	NO	—	Unique issue number, e.g. ISS-2026-00001
production_order_id	BIGINT	NO	—	FK to Production Order; identifies the production order requesting the materials
issue_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date and time materials are issued
issued_by	BIGINT	NO	—	FK to User/Employee; storekeeper who issued the materials
received_by	BIGINT	NO	—	FK to User/Employee; production supervisor who received the materials
status	VARCHAR(20)	NO	'DRAFT'	Issue status: DRAFT, APPROVED, POSTED, CANCELLED
remarks	TEXT	YES	—	General remarks about the material issue
created_by	BIGINT	NO	—	User who created the issue
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Record creation date/time
posted_by	BIGINT	YES	—	User who posted the issue to inventory
posted_at	TIMESTAMP	YES	—	Date/time inventory was updated
created_by	BIGINT	NO	—	FK to User; user who created the transfer
created_at	TIMESTAMP	NO	CURRENT_TIME	Date/time transfer was created
Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Updated_by	TEXT	YES		The user who updated this record
operating_unit_id	BIGINT	NO	-	FK to operating units; identifies the factory where the transaction occurs

Issue Line

Column	PostgreSQL Data Type	NULL?	Default	Purpose
issue_line_id	BIGSERIAL	NO	Identity	Primary key; unique issue line
issue_id	BIGINT	NO	—	FK to Issue Header
material_id	BIGINT	NO	—	FK to Item/Material Master; material issued to production
batch_id	BIGINT	NO	—	FK to Batch/Lot Master; batch issued
uom_id	BIGINT	NO	—	FK to UOM Master
requested_qty	NUMERIC(18,3)	NO	0	Quantity requested by production
issued_qty	NUMERIC(18,3)	NO	0	Actual quantity issued by the storekeeper
remarks	TEXT	YES	—	Line-specific remarks
created_by	BIGINT	NO	—	FK to User; user who created the transfer
created_at	TIMESTAMP	NO	CURRENT_TIME	Date/time transfer was created
Updated_at	TIMESTAMP	YES		Date and time the ledger record was updated
Updated_by	TEXT	YES		The user who updated this record

10. Physical Count

Information Header

Column	Data Type	NULL?	Default	Purpose
count_id	BIGSERIAL	NO	Identity	Primary key; unique physical count record ID
count_no	VARCHAR(30)	NO	—	Unique physical count number, e.g. CNT-2026-00001
warehouse_id	BIGINT	NO	—	FK to Warehouses; warehouse being physically counted
storage_location_id	BIGINT	YES	—	FK to Storage Location/Store; specific location being counted
count_date	TIMESTAMP	NO	CURRENT_TIME	Date and time the physical count was performed
counted_by	BIGINT	NO	—	FK to Employee/User; person or team responsible for counting
remark	TEXT	YES	—	Additional comments or observations
status	VARCHAR(20)	NO	'DRAFT'	Count status: DRAFT, COUNTED, REVIEWED, APPROVED, POSTED, CANCELLED
created_by	BIGINT	NO	—	FK to User; user who created the count

created_date	TIMESTAMP	NO	CURRENT_TIME	Record creation date/time
updated_by	BIGINT	YES	—	FK to User; last user who updated the record
updated_date	TIMESTAMP	YES	CURRENT_TIME	Last update date/time
operating_unit_id	BIGINT	NO		

Counted Item

Column	Data Type	NULL?	Default	Purpose
counted_item_id	BIGSERIAL	NO	Identity	Primary key; unique physical count line ID
count_id	BIGINT	NO	—	FK to count_information_header.count_id; identifies the physical count
material_id	BIGINT	NO	—	FK to Item/Material Master; identifies the counted item
item_code	VARCHAR(30)	NO	—	Item/material code; normally retrieved from Material Master
uom_id	BIGINT	NO	—	FK to UOM Master; unit of measure
bin_id	BIGINT	YES	—	FK to Bin Master; physical bin where the item was counted
batch_id	BIGINT	YES	—	FK to Batch/Lot Master; important for wheat and finished flour traceability
system_qty	NUMERIC(18,3)	NO	0	Quantity recorded in the ERP before physical counting
physical_qty	NUMERIC(18,3)	NO	0	Actual quantity physically counted
variance	NUMERIC(18,3)	NO	0	Difference between physical quantity and system quantity
status	VARCHAR(20)	NO	'PENDING'	Line status: PENDING, MATCHED, SHORTAGE, SURPLUS, APPROVED, ADJUSTED
remark	TEXT	YES	—	Explanation or comments about the count/variance
created_by	BIGINT	NO	—	FK to User; user who created the count line
created_date	TIMESTAMP	NO	CURRENT_TIMESTAMP	Record creation date/time
updated_by	BIGINT	YES	—	FK to User; last user who updated the record
updated_date	TIMESTAMP	YES	CURRENT_TIMESTAMP	Last update date/time

8. Inventory Adjustment

Column	Data Type	NULL?	Default	Purpose
adjustment_id	BIGSERIAL	NO	Identity	Primary key; unique inventory adjustment ID
adjustment_number	VARCHAR(30)	NO	—	Unique adjustment number, e.g. ADJ-2026-00001
count_id	BIGINT	YES	—	FK to Physical Count Header; identifies the count that caused the adjustment
material_id	BIGINT	NO	—	FK to Item/Material Master; material being adjusted
batch_id	BIGINT	YES	—	FK to Batch/Lot Master; batch being adjusted
bin_id	BIGINT	YES	—	FK to Bin Master; physical location of the adjustment
system_qty	NUMERIC(18,3)	NO	0	Quantity recorded in ERP before adjustment
physical_qty	NUMERIC(18,3)	NO	0	Quantity physically counted
variance_qty	NUMERIC(18,3)	NO	0	Difference between physical quantity and system quantity
reason	VARCHAR(100)	NO	—	Reason for adjustment, e.g. DAMAGE, LOSS, CORRECTION, COUNT_VARIANCE
remarks	TEXT	YES	—	Additional explanation for the adjustment
approved_by	BIGINT	YES	—	FK to User/Employee; manager who approved the adjustment
approved_at	TIMESTAMP	YES	—	Date/time adjustment was approved
status	VARCHAR(20)	NO	'PENDING'	PENDING, APPROVED, POSTED, REJECTED, CANCELLED
created_by	BIGINT	NO	—	User who created the adjustment
created_at	TIMESTAMP	NO	CURRENT_TIME	Date/time adjustment was created
posted_by	BIGINT	YES	—	User who posted the adjustment to inventory
posted_at	TIMESTAMP	YES	—	Date/time adjustment was posted
Oprating_unit_id	BIGINT	NO		
updated_by	BIGINT	YES	—	FK to User; last user who updated the record
updated_date	TIMESTAMP	YES	CURRENT_TIME	Last update date/time

Payable

1. Supplier Invoice Registration

HEADER

Column	Data Type	NULL?	Default	Purpose
supplier_invoice_id	BIGSERIAL	NO	Identity	Primary key; unique supplier invoice record
invoice_no	VARCHAR(50)	NO	—	Supplier invoice number/document number
invoice_date	DATE	NO	—	Date shown on the supplier invoice
supplier_id	BIGINT	NO	—	FK to Supplier Master; identifies the supplier
Operatining_unit_id	BIGINT	NO		
supplier_ref	VARCHAR(100)	YES	—	Supplier's reference number or document reference
invoice_type	VARCHAR(30)	NO	—	Invoice type, e.g. GOODS, SERVICE, CREDIT_NOTE, DEBIT_NOTE
currency	VARCHAR(3)	NO	'ETB'	Currency of the supplier invoice, e.g. ETB, USD
purchase_order_ref	VARCHAR(50)	YES	—	Purchase Order reference
goods_receipt_ref	VARCHAR(50)	YES	—	GRN/Goods Receipt reference
delivery_note_ref	VARCHAR(50)	YES	—	Supplier Delivery Note reference
supplier_invoice_attachment	VARCHAR(500)	YES	—	File path/storage reference for the supplier invoice attachment
other_document_attachment	VARCHAR(500)	YES	—	File path/storage reference for supporting document
other_document_attachment_1	VARCHAR(500)	YES	—	File path/storage reference for an additional supporting document
approval_status	VARCHAR(20)	NO	'PENDING'	Approval status: PENDING, APPROVED, REJECTED, CANCELLED
Invoice_amount	NUMERIC(18,4)	NO		
approval_id	BIGINT	YES	—	FK to Approval/Approval Workflow record
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date/time the invoice record was created
created_by	BIGINT	NO	—	FK to User; user who created the invoice
updated_at	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date/time the invoice was last updated
updated_by	BIGINT	YES	—	FK to User; user who last updated the invoice

LINE

Column	Data Type	NULL?	Default	Purpose
supplier_invoice_item_id	BIGSERIAL	NO	Identity	Primary key; unique supplier invoice line ID
supplier_invoice_id	BIGINT	NO	—	FK to Supplier Invoice Header; identifies the parent invoice
material_id	BIGINT	NO	—	FK to Item/Material Master; identifies the invoiced item
invoice_item_code	VARCHAR(30)	NO	—	Item/material code appearing on the supplier invoice
qty	NUMERIC(18,3)	NO	0	Quantity invoiced
uom_id	BIGINT	NO	—	FK to UOM Master; unit of measure for the quantity
unit_price	NUMERIC(18,4)	NO	0	Price per unit before tax
tax	NUMERIC(18,4)	NO	0	Tax amount applicable to the invoice line
line_amount	NUMERIC(18,4)	NO	0	Quantity × unit price before tax
total_amount	NUMERIC(18,4)	NO	0	Line amount plus applicable tax
created_at	TIMESTAMP	NO	CURRENT_TIMESTAMP	Date/time the invoice line was created
created_by	BIGINT	NO	—	FK to User; user who created the line
updated_at	TIMESTAMP	YES	CURRENT_TIMESTAMP	Date/time the invoice line was last updated
updated_by	BIGINT	YES	—	FK to User; user who last updated the line

2. Supplier Payment Request

Field Name	Data Type	Size	Null	Description
payment_request_no	VARCHAR	30	Yes	Unique payment request number generated by the system. Example: SPR-2026-00001.
invoice_no_system	VARCHAR	30	Yes	Reference to the supplier invoice number/ID recorded in the system. Used to link the payment request to the supplier invoice.
invoice_ref_doc	VARCHAR	100	Yes	Supplier's original invoice/reference document number appearing on the submitted invoice.
supplier_id	BIGINT	—	Yes	Unique identifier of the supplier requesting payment. Foreign key to the **Supplier Master** table.
amount_to_pay	DECIMAL	18,2	Yes	Amount requested for payment against the supplier invoice. Should not normally exceed the approved outstanding invoice balance.
requested_date	DATE	—	Yes	Date on which the supplier payment request was created/submitted.
approval_status	VARCHAR	20	Yes	Current approval status of the payment request. Examples: DRAFT, PENDING, APPROVED, REJECTED, CANCELLED, PAID.
approval_id	BIGINT	—	No	Reference to the approval workflow/approval record associated with this payment request. Foreign key to the **Approval** table.
operating_units_id	BIGINT	—	Yes	Identifies the operating unit responsible for the transaction. Foreign key to the **Operating Unit** master table.
created_at	TIMESTAMP	—	Yes	Date and time when the payment request record was created.
created_by	BIGINT	—	Yes	User ID of the person who created the payment request. Foreign key to the **User** table.
updated_at	TIMESTAMP	—	No	Date and time when the payment request was last modified.
updated_by	BIGINT	—	No	User ID of the person who last modified the payment request.
payment_request_id	BIGINT	19	Yes	Primary key of the payment request.
currency_id	BIGINT	19	Yes	Foreign key identifying the payment currency.
invoice_amount	DECIMAL	18,2	Yes	Total amount of the supplier invoice.
payment_reason	VARCHAR	500	No	Reason or justification for requesting the payment.
attachment_path	VARCHAR	100	No	File path, URL, or document reference for supporting attachments.
remarks	VARCHAR	1,000	No	Additional comments or notes about the payment request.
rejection_reason	VARCHAR	1,000	No	Explanation provided when the payment request is rejected.

3. Supplier Payment Execution

Field Name	Data Type	Size	Required	Description
payment_execution_id	BIGINT	19	Yes	Primary key uniquely identifying the payment execution record.
payment_request_no	VARCHAR	30	Yes	Reference to the approved Supplier Payment Request.
payment_no	VARCHAR	30	Yes	Unique payment transaction number generated by the system. Example: PAY-2026-00001.
payment_date	DATE	—	Yes	Date on which the supplier payment was executed.
supplier_id	BIGINT	19	Yes	Foreign key identifying the supplier receiving the payment.
payment_method_id	VARCHAR	30	Yes	Method used to make the payment. Examples: Bank Transfer, Cheque, Cash, Online Transfer.
currency_id	VARCHAR	3	Yes	Currency code of the payment. Example: ETB, USD, EUR.
payment_amount	DECIMAL	18,2	Yes	Actual amount paid to the supplier.
amount_in_words	VARCHAR	500	Yes	Payment amount expressed in words. Example: Five Hundred Thousand Birr Only.
reference	VARCHAR	100	No	General payment reference or supporting reference number.
bank_account_id	VARCHAR	100	No	Supplier bank account or selected company bank account used for the transaction.
transaction_ref	VARCHAR	100	No	Bank/payment transaction reference generated by the bank or payment service.
upload_deposit_slip	VARCHAR	500	No	File path, URL, or document reference for the bank deposit/payment slip.
remaining_amount	DECIMAL	18,2	No	Remaining unpaid amount after this payment execution.
operating_units_id	BIGINT	19	Yes	Foreign key identifying the operating unit responsible for the payment.
created_at	TIMESTAMP	—	Yes	Date and time the payment execution record was created.
created_by	BIGINT	19	Yes	User ID of the person who created the payment execution record.
updated_at	TIMESTAMP	—	No	Date and time the payment record was last updated.
updated_by	BIGINT	19	No	User ID of the person who last updated the payment record.